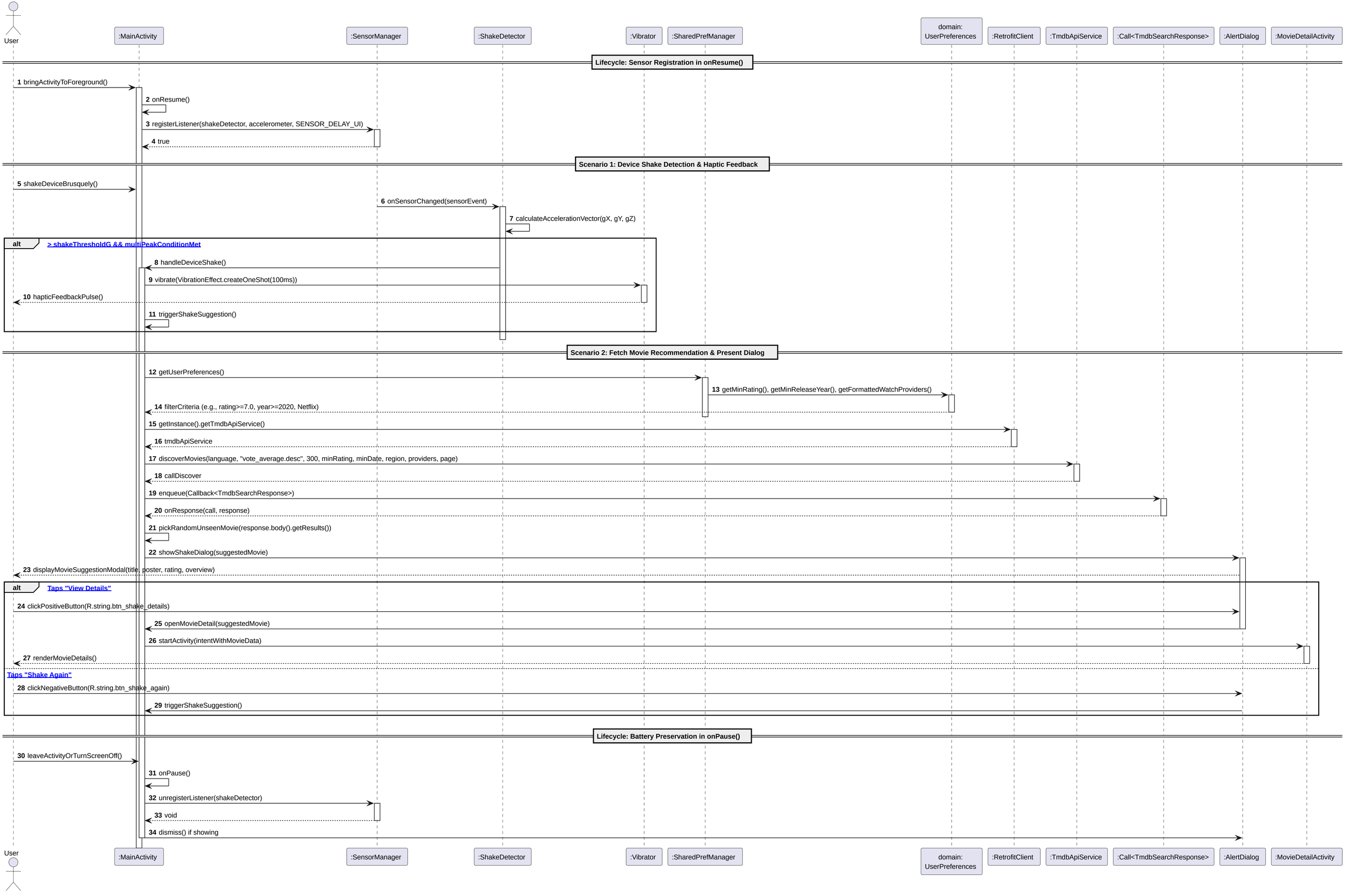
User
:MainActivity
:SensorManager
:ShakeDetector
:Vibrator
:SharedPrefManager
domain: UserPreferences
:RetrofitClient
:TmdbApiService
:Call<TmdbSearchResponse>
:AlertDialog
:MovieDetailActivity
Lifecycle: Sensor Registration in onResume()
1 bringActivityToForeground()
2 onResume()
3 registerListener(shakeDetector, accelerometer, SENSOR_DELAY_UI)
4 true
Scenario 1: Device Shake Detection & Haptic Feedback
5 shakeDeviceBrusquely()
6 onSensorChanged(sensorEvent)
7 calculateAccelerationVector(gX, gY, gZ)
alt
> shakeThresholdG && multiPeakConditionMet
8 handleDeviceShake()
9 vibrate(VibrationEffect.createOneShot(100ms))
10 hapticFeedbackPulse()
11 triggerShakeSuggestion()
Scenario 2: Fetch Movie Recommendation & Present Dialog
12 getUserPreferences()
13 getMinRating(), getMinReleaseYear(), getFormattedWatchProviders()
14 filterCriteria (e.g., rating>=7.0, year>=2020, Netflix)
15 getInstance().getTmdbApiService()
16 tmdbApiService
17 discoverMovies(language, "vote_average.desc", 300, minRating, minDate, region, providers, page)
18 callDiscover
19 enqueue(Callback<TmdbSearchResponse>)
20 onResponse(call, response)
21 pickRandomUnseenMovie(response.body().getResults())
22 showShakeDialog(suggestedMovie)
23 displayMovieSuggestionModal(title, poster, rating, overview)
alt
Taps "View Details"
24 clickPositiveButton(R.string.btn_shake_details)
25 openMovieDetail(suggestedMovie)
26 startActivity(intentWithMovieData)
27 renderMovieDetails()
Taps "Shake Again"
28 clickNegativeButton(R.string.btn_shake_again)
29 triggerShakeSuggestion()
Lifecycle: Battery Preservation in onPause()
30 leaveActivityOrTurnScreenOff()
31 onPause()
32 unregisterListener(shakeDetector)
33 void
34 dismiss() if showing